

Arthur R. C. McCray

Postdoctoral Scholar, Stanford University

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SUMMARY

Research scientist working at the intersection of machine learning and electron microscopy. Builds self-supervised and physics-informed methods for inverse problems in scientific imaging: deep generative priors, implicit neural representations, and differentiable forward models. Ships them as open-source tools used across the field. Recent work produced $>10\times$ faster and 40% higher-resolution ptychographic reconstructions, secured \$1.2M in funding from Toyota Research Institute, and is being transferred into semiconductor metrology workflows.

TECHNICAL SKILLS

Machine learning: self-supervised learning; inverse problems; implicit neural representations; deep generative priors and diffusion models; vision transformers and CNNs; differentiable programming; iterative reconstruction and optimization; uncertainty and robustness evaluation.

Software: Python (NumPy/SciPy/scikit-learn), PyTorch, CUDA and JAX (working knowledge); package design and distribution, testing and CI, documentation, open-source maintenance; git; Mathematica; LabVIEW.

Systems: GPU training and profiling; distributed training at medium scale (up to 128 nodes); data pipelines for TB-scale imaging datasets; HPC clusters (NERSC).

Simulation: multislice electron scattering; micromagnetics (Mumax, OOMMF).

Microscopy and characterization: Lorentz TEM, HR(S)TEM, 4D-STEM, electron diffraction, cryo-TEM (LN₂ and LHe), in-situ biasing and heating, EDS mapping, sample preparation; MPMS/SQUID, MOKE, XRR, SEM, AFM.

Nanofabrication: FIB patterning and lift-out, photolithography, sputtering, 2D-material exfoliation.

RESEARCH EXPERIENCE

Stanford University, Stanford, CA

Sept 2024 – Present

Postdoctoral Scholar: Ophus Group

- Proposed and demonstrated a self-supervised deep-prior framework for solving inverse problems in (S)TEM and established it as a reusable approach now underpinning projects in ptychography, tomography, and crystallography. Supervise extensions of the method by three graduate students and two postdoctoral researchers at Stanford, with active collaborations at SLAC, UC Berkeley, and the Technion.
- Initiated and led the first application of this framework to electron ptychography, achieving a $>10\times$ speedup and 40% resolution improvement over state-of-the-art ML and iterative baselines.

- Led the technical contributions behind a \$1.2M award from Toyota Research Institute for Deep Material Priors, a program building physics-aware foundation models for materials characterization.
- Developed whole-pattern-fitting methods for improved strain mapping, now being adopted for semiconductor metrology workflows at Samsung, and supervise their ongoing development.
- Created quantEM, an open-source library for quantitative electron microscopy, and lead its ptychography and machine learning development.
- Provide day-to-day technical leadership across these projects: scoping the work, reviewing code and experiments, and guiding model design and evaluation.

Lawrence Berkeley National Laboratory, Berkeley, CA

Feb 2024 – Sept 2024

Postdoctoral Scholar: National Center for Electron Microscopy (NCEM)

- Developed machine learning methods for 4D-STEM analysis in the group of Colin Ophus; continued the same research program upon the group's move to Stanford.

Argonne National Laboratory / Northwestern University, IL

2018 – 2023

Graduate Research Assistant: Petford-Long and Phatak Groups

- Built a simulation and training pipeline for Lorentz TEM neural networks, enabling large-scale statistical studies of spin textures.
- Developed a self-supervised method for quantitative magnetic phase imaging from a single image, replacing the conventional through-focal series.
- Authored PyLorentz, an open-source Python package for Lorentz TEM simulation and phase reconstruction that is now widely used in the magnetic-microscopy community.
- Designed and ran cryogenic and in-situ TEM experiments on van der Waals magnets.
- Collaborated across 10+ groups at national labs and universities, contributing analysis and imaging expertise to publications.

OPEN SOURCE

quantEM: core developer and package creator

github.com/electronmicroscopy/quantem

- Open-source Python library for quantitative electron microscopy (50+ GitHub stars).
- Developed the ptychography and machine learning modules; contributed core infrastructure (data structures, visualization, utilities, configuration) and tutorials.

PyLorentz: sole developer and maintainer

github.com/PyLorentz/PyLorentz

- Widely adopted Python codebase for Lorentz TEM simulation, phase reconstruction, and ML-based analysis (35+ GitHub stars).
- Sole author and maintainer since 2020: architecture, documentation, releases, and user support.

Additional projects at github.com/arthurmccray

EDUCATION

Northwestern University, Evanston, IL

2018 – 2023

- **Ph.D., Applied Physics.** Thesis: *Observing and Understanding Complex Magnetic Spin Textures with Lorentz Microscopy* (Advisor: Amanda Petford-Long).

Carleton College, Northfield, MN

2018

- **B.A., Physics, *cum laude*.** Graduated with Distinction for the Senior Integrative Exercise.

PUBLICATIONS

Full and current list at [Google Scholar](https://scholar.google.com/citations?user=...). * denotes equal contribution.

FIRST-AUTHOR PUBLICATIONS

Arthur R. C. McCray, Stephanie Ribet, Georgios Varanavides, and Colin Ophus. “Deep generative priors for robust and efficient electron ptychography.” arXiv:2511.07795 (2025).

Arthur R. C. McCray, Stephanie Ribet, Georgios Varanavides, and Colin Ophus. “Accelerating Iterative Ptychography with an Integrated Neural Network.” *Journal of Microscopy*, 300, 180–190. (2025).

Arthur R. C. McCray, Alec Bender, Amanda Petford-Long, and Charudatta Phatak. “Simulation-Trained Machine Learning Models for Lorentz Transmission Electron Microscopy.” *APL Machine Learning* 2, 026120 (2024).

Arthur R. C. McCray, Tao Zhou, Saugat Kandel, Amanda Petford-Long, Mathew J. Cherukara, and Charudatta Phatak. “AI-Enabled Lorentz Microscopy for Quantitative Imaging of Nanoscale Magnetic Spin Textures.” *npj Computational Materials* 10, 111 (2024).

Arthur R. C. McCray, Yue Li, Eric Qian, Yi Li, Wei Wang, Zhengjie Huang, Xuedan Ma, Yuzi Liu, Duck Young Chung, Mercouri G. Kanatzidis, Amanda K. Petford-Long, and Charudatta Phatak. “Direct Observation of Magnetic Bubble Lattices and Magnetoelastic Effects in van der Waals Cr₂Ge₂Te.” *Advanced Functional Materials* 2214203 (2023).

Arthur R. C. McCray*, Yue Li*, Rabindra Basnet, Krishna Pandey, Jin Hu, Daniel P. Phelan, Xuedan Ma, Amanda K. Petford-Long, and Charudatta Phatak. “Thermal Hysteresis and Ordering Behavior of Magnetic Skyrmion Lattices.” *Nano Letters* 22, 7804–7810 (2022).

Arthur R. C. McCray, Tim Cote, Yue Li, Amanda K. Petford-Long, and Charudatta Phatak. “Understanding Complex Magnetic Spin Textures with Simulation-Assisted Lorentz Transmission Electron Microscopy.” *Physical Review Applied* 15, 044025 (2021). [Editors’ Suggestion]

Arthur R. C. McCray*, Benjamin H. Savitzky*, Kevin Whitham, Tobias Hanrath, and Lena F. Kourkoutis. “Orientational Disorder in Epitaxially Connected Quantum Dot Solids.” *ACS Nano* 13, 11460–11468 (2019).

CO-AUTHORED PUBLICATIONS

Stephanie M. Ribet, Mohsen Danaie, Willem P. M. de Kleijne, **Arthur R. C. McCray**, Frederick Allars, Christopher S. Allen, Colin Ophus, and Georgios Varnavides. “Real-space overlap is not enough: ambiguity in nanobeam iterative ptychography.” arXiv:2608.30359 (2026).

Jared Coles, **Arthur R. C. McCray**, Yue Li, Bryan T. Fichera, Yan Wu, Yiqing Hao, Daniel Phelan, Yue Cao, Raymond Osborn, Charudatta Phatak, Stephan Rosenkranz, and Yu Li. “Model-free Detection of Scattering and Imaging Data using Escort-Weighted Shannon Entropy and Divergence Matrices.” *Physical Review Research*, 8, 3, 033235 (2026).

- Nicholas Marchese, **Arthur R. C. McCray**, Yael Tsarfati, Karen Bustillo, Adam Marks, Alberto Salleo, and Colin Ophus. “Mapping Order in Semicrystalline Polymers using Machine Learning of Nanobeam Electron Diffraction.” arXiv:2607.16570 (2026).
- Serin Lee, Stephanie M. Ribet, **Arthur R. C. McCray**, Andrew Barnum, Jennifer A. Dionne, and Colin Ophus. “Unsupervised Segmentation and Clustering Workflow for Efficient Processing of 4D-STEM and 5D-STEM Data.” *Microscopy and Microanalysis* 32, 2, ozag044 (2026).
- Georgios Varnavides, Julie M. Bekkevold, Stephanie M. Ribet, **Arthur R. C. McCray**, Mary C. Scott, Lewys Jones, and Colin Ophus. “Beyond Contrast Transfer: Spectral SNR as a Finite-Dose Metric for STEM Phase Retrieval.” *Microscopy and Microanalysis* 32, 2, ozag005 (2026).
- Anvesh Nathani, **Arthur R. C. McCray**, Yingtao Liu, Hanping Ding, Pejman Kazempoor, Shuozhi Xu, Colin Ophus, and Iman Ghamarian. “Accelerating Electron Diffraction Analysis Using Graph Neural Networks and Attention Mechanisms.” *npj Computational Materials* 12, 56 (2026).
- Cedric Lim, Corneel Casert, **Arthur R. C. McCray**, Serin Lee, Andrew Barnum, Jennifer Dionne, and Colin Ophus. “Missing Wedge Inpainting and Joint Alignment in Electron Tomography through Implicit Neural Representations.” arXiv:2512.08113 (2025).
- N. M. Singh, T. Chien, **Arthur R. C. McCray**, C. Ophus, and L. Waller. “A Gaussian Parameterization for Direct Atomic Structure Identification in Electron Tomography.” *IEEE International Conference on Computational Photography (ICCP)* 1–10 (2025).
- Myeongjin Jang, Sol Lee, Fernando Cantos-Prieto, Ivona Košić, Yue Li, **Arthur R. C. McCray**, Min-Hyoung Jung, Jun-Yeong Yoon, Loukya Boddapati, Francis Leonard Deepak, Hu Young Jeong, Charudatta Phatak, Elton Santos, Efrén Navarro-Moratalla, and Kwanpyo Kim. “Direct Observation of Twisted Stacking Domains in the van der Waals Magnet CrI₃.” *Nature Communications* 15, 5925 (2024).
- John Fullerton, **Arthur R. C. McCray**, Amanda K. Petford-Long, and Charudatta Phatak. “Understanding the Effect of Curvature on the Magnetization Reversal of Three-Dimensional Nanohelices.” *Nano Letters* 24, 2481-2487 (2024).
- Sergey Grebenchuk, Conor McKeever, Magdalena Grzeszczyk, Zhaolong Chen, Makars Šiškins, **Arthur R. C. McCray**, Yue Li, Amanda Petford-Long, Charudatta Phatak, Duan Ruihuan, Liu Zheng, Kostya Novoselov, Elton Santos, and Maciej Koperski. “Topological Spin Textures in an Insulating van der Waals Ferromagnet.” *Advanced Materials* 36, 2311949 (2024).
- Zhengjie Huang, **Arthur R. C. McCray**, Yue Li, Darien J. Morrow, Eric K. Qian, Duck Young Chung, Mercouri G. Kanatzidis, Charudatta Phatak, and Xuedan Ma. “Raman Shifts in Two-Dimensional van der Waals Magnets Reveal Magnetic Texture Evolution.” *Nano Letters* 24, 1531-1538 (2024).
- Yue Li, **Arthur R. C. McCray**, Wei Wang, Rabindra Basnet, Krishna Pandey, Xuedan Ma, Jin Hu, Amanda Petford-Long, and Charudatta Phatak. “Control of Skyrmion Lattice Order in the van der Waals ferromagnet Fe₃GeTe₂.” *Spintronics* 12656, 30 (2023).

Vuk Brajuskovic, **Arthur R. C. McCray**, Yuepeng Zhang, and Charudatta Phatak. “In situ observation of the magnetization configuration and reversal in cylindrical nanowires.” APL Materials 10, 081109 (2022).

Yue Li, Frank Barrows, **Arthur R. C. McCray**, Tim Cote, David Friedman, Ralu N. S. Divan, Amanda K. Petford-Long, and Charudatta Phatak. “Geometric Control of Emergent Antiferromagnetic Order in Coupled Artificial Spin Ices.” Cell Reports Physical Science 3, 100846 (2022).

Yue Li, Rabindra Basnet, Krishna Pandey, Jin Hu, Wei Wang, Xuedan Ma, **Arthur R. C. McCray**, Amanda K. Petford-Long, and Charudatta Phatak. “Field Dependent Magnetic Behavior in van der Waals Fe₃GeTe₂.” JOM: The Journal of Minerals, Metals & Materials Society 74, 2310-2318 (2022).

PRESENTATIONS

Invited: “Deep Priors and Self-Supervised Learning for Electron Ptychography and Beyond.” Microscopy and Microanalysis Conference, Milwaukee, WI, 2026.

“Improving Iterative Ptychography with Generative Priors.” Materials Research Society Spring Conference, Honolulu, HI, 2026.

Invited: “Improving Iterative Ptychography with Machine Learning.” Advanced Photon Source ptychography study group, 2025.

“Improving Iterative Ptychographic Reconstructions with Generative Neural Networks.” Microscopy and Microanalysis Conference, Salt Lake City, UT, 2025.

“Deep Learning Model for Obtaining Spatially Resolved Pair Distribution Functions from 4D-STEM Data.” Microscopy and Microanalysis Conference, Salt Lake City, UT, 2025.

Invited: “Solving Inverse Problems in TEM with Machine Learning.” SLAC Battery Center Seminar Series, 2025.

“ML-Enabled Single Image Magnetic Phase Reconstruction for Lorentz Transmission Electron Microscopy.” Microscopy and Microanalysis Conference, Cleveland, OH, 2024.

“3D Magnetization Reconstruction for Lorentz Microscopy using Differential Programming.” Microscopy and Microanalysis Conference, Minneapolis, MN, 2023.

“Behavior of Bubble Lattices in Cr₂Ge₂Te₆.” 67th MMM conference, Minneapolis, MN, Oct 31 – Nov 4, 2022.

Invited: “Evolution of Skyrmion Lattice Order in the van der Waals Ferromagnet Fe₃GeTe₂.” Microscopy and Microanalysis, Portland, OR, 2022.

“Simulation-Trained Machine Learning Models for Lorentz Microscopy.” Microscopy and Microanalysis, Portland, OR, 2022.

Northwestern University Applied Physics Seminar: “Understanding Complex Magnetic Structures with Simulation-Assisted Lorentz Transmission Electron Microscopy.” Northwestern University, 2021.

Invited: “Convolutional Neural Networks for Image Processing.” Biomedical engineering seminar series, University of Illinois Urbana Champaign, 2021.

POSTER PRESENTATIONS

- “Accelerating Ptychographic Phase Reconstructions with ML-Enabled Phase Unwrapping.” Microscopy and Microanalysis Conference, Cleveland, OH, 2024.
- “Control of Magnetic Skyrmions in an Exchange Biased van der Waals Ferromagnet.” Microscopy and Microanalysis Conference, Minneapolis, MN, 2023.
- “Cryogenic Lorentz Transmission Electron Microscopy of van der Waals Materials.” 2022 Joint Meeting of IEEE Magnetics Society & Nanotechnology Council, IEEE Chicago Section.
- “Tailoring Magnetic Skyrmions in Multilayers by Anisotropy Engineering.” 2019 Joint Meeting of IEEE Magnetics Society & Nanotechnology Council, IEEE Chicago Section.
- “Orientational Disorder in Epitaxially Connected Quantum Dot Solids.” Microscopy and Microanalysis Conference, St. Louis, MO, 2017.

HONORS AND AWARDS

Postdoctoral Scholar Award, Microscopy and Microanalysis Conference	2025
Student Scholar Award, Microscopy and Microanalysis Conference	2022
Editors’ Suggestion, Physical Review Applied	2021
Best Poster, IEEE Magnetics Society & Nanotechnology Council, Chicago Section	2019
Distinction for Senior Integrative Exercise, Carleton College	2018

RESEARCH INTERESTS

- **Physics-informed and self-supervised machine learning:** algorithms constrained by physical models including electron scattering and magnetic energy terms.
- **ML-enabled solutions to inverse problems:** model parameterization and priors, implicit neural representations, splatting.
- **Foundation models for materials:** deep material priors trained from first-principles simulation.
- **Neural network interpretability:** understanding physical systems through latent representations and understanding how networks solve problems with known solutions.
- **Nanoscale magnetism:** topologically protected spin textures and van der Waals ferromagnets.
- **In-situ (S)TEM:** evolution and response of functional materials under applied stimuli.

PROFESSIONAL ACTIVITIES

Journal referee for npj Computational Materials; APL Machine Learning; Advanced Functional Materials; Microscopy & Microanalysis; Ultramicroscopy; Nanoscale Horizons; Journal of Applied Physics; Scientific Reports; IEEE Transactions on Magnetics; Digital Discovery.

TEACHING AND MENTORING

- Graduate student mentor: Stanford University** 2024 – Present
- Day-to-day technical supervision of graduate students on ML and microscopy projects.

Guest Lecturer: Stanford University

Fall 2024, Spring 2025

- MatSci 327: TEM Analysis and Simulation.

Graduate Teaching Assistant: Northwestern University

2019, 2022

- Physics and Materials Science & Engineering departments.

Undergraduate Teaching Assistant: Carleton College

2016 – 2018

- Physics department.

OUTREACH

Scientist of the Month

2020 – Present

- Volunteer scientist at the Lakeside School and the Evergreen School, Seattle.

Stanford Science Pen-Pals

2024 – Present

Mentorship Opportunities for Research Engagement (MORE)

2019 – 2023

- Northwestern University: board member and REACH coordinator.

Northwestern Prison Education Program

2020 – 2023

Future City Mentor: Saint Athanasius Middle School

2018 – 2020

Invited outreach talk: Metea Valley High School

2022

- “High School to Physics PhD,” [Open Gears Seminar](#).